Problem Set 3

Applied Stats/Quant Methods 1

Due: November 12, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
# read in data
inc.sub <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/incumbents_subset.csv")

#question1 1 Run a regression where the outcome variable is voteshare and the explanatory variable is difflog
regression_question1 <- lm(voteshare~difflog, data = inc.sub)
# get summary of model
summary(regression_question1)
```

```
Call:

lm(formula = voteshare ~ difflog, data = inc.sub)


Residuals:

Min       1Q   Median       3Q      Max

-0.26832 -0.05345 -0.00377  0.04780  0.32749


Coefficients:

Estimate Std. Error t value Pr(>|t|)

(Intercept) 0.579031   0.002251  257.19   <2e-16 ***

difflog     0.041666   0.000968   43.04   <2e-16 ***

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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

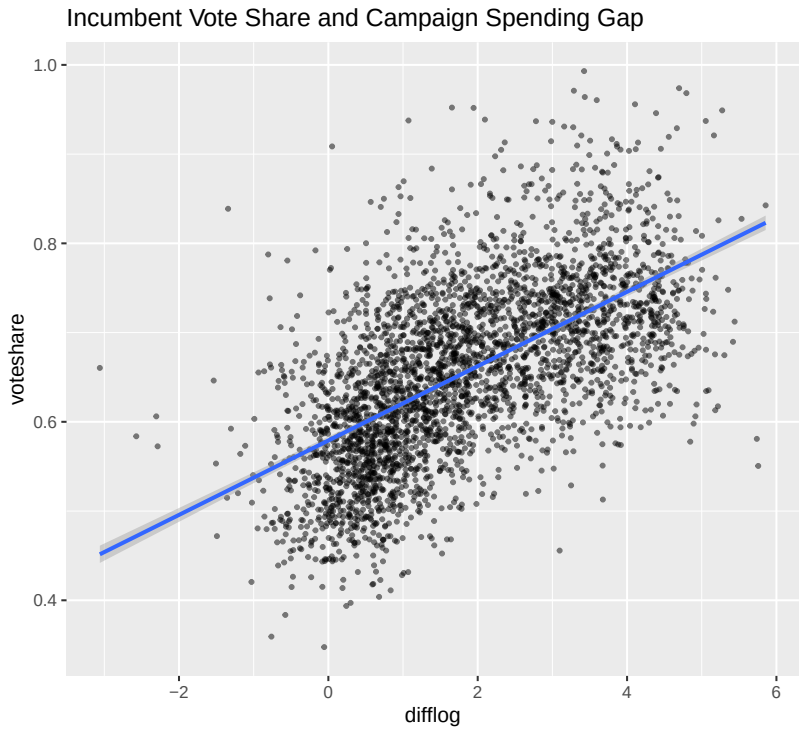

Residual standard error: 0.07867 on 3191 degrees of freedom

Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671

F-statistic:  1853 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

```r
#question1 2
library(ggplot2)
scattor_question1 <- ggplot(inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point(size = 0.8, alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "difflog", y = "voteshare", title = "Incumbent Vote Share and
    Campaign Spending Gap")
ggsave("scattor_question1.pdf")
```



3. Save the residuals of the model in a separate object.

```r
#question1 3
res_question1 <- resid(regression_question1)
```

4. Write the prediction equation.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.579031   0.002251  257.19   <2e-16 ***
difflog     0.041666   0.000968   43.04   <2e-16 ***
```

From the summary of the regression model, we can find that the intercept is 0.579 and the slope is 0.042. We can write the prediction equation as:

$$\hat{y} = 0.042x + 0.579$$

where y is `voteshare` and x is `difflog`. On average, a one-unit increase in `difflog` is associated with a 0.042 unit increase in `voteshare`. This coefficient is statistically significant at the $\alpha = 0.05$ level ($p < 0.001$). When `difflog` is 0, the predicted `voteshare` is 0.579 (about 57.9%).

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
regression_question2 <- lm(presvote~difflog, data = inc.sub)
#get a summary of model
summary(regression_question2)
```

```
Call:
lm(formula = presvote ~ difflog, data = inc.sub)

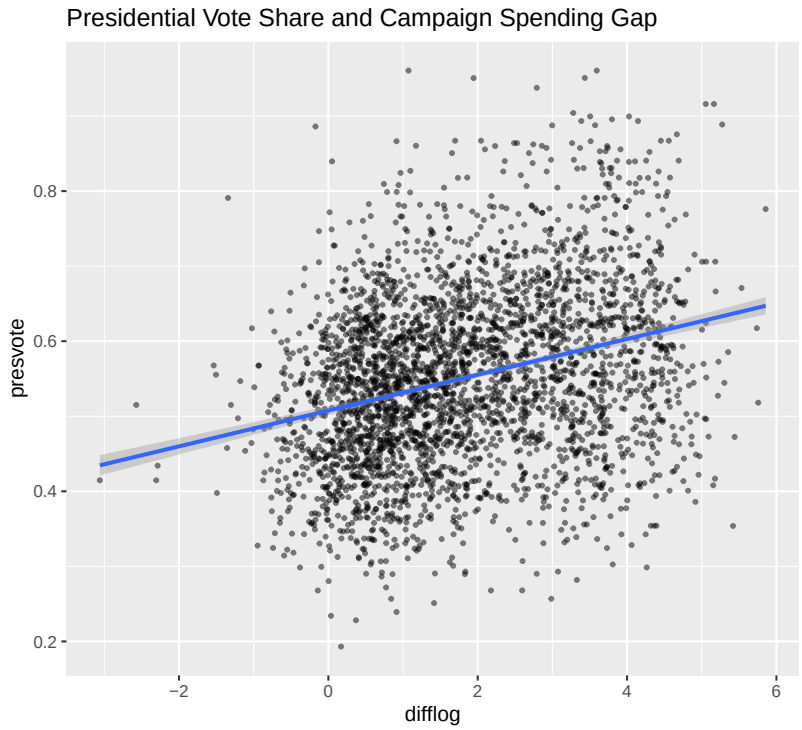
Residuals:
Min       1Q   Median       3Q      Max
-0.32196 -0.07407 -0.00102  0.07151  0.42743

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.507583   0.003161  160.60   <2e-16 ***
difflog     0.023837   0.001359   17.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1104 on 3191 degrees of freedom
Multiple R-squared:  0.08795, Adjusted R-squared:  0.08767
F-statistic: 307.7 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

```
scattor_question2 <- ggplot(inc.sub, aes(x = difflog, y = presvote)) +
  geom_point(size = 0.8, alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "difflog", y = "presvote", title = "Presidential Vote Share and
    Campaign Spending Gap")
#scattor_question2
ggsave("scattor_question2.pdf")
```



3. Save the residuals of the model in a separate object.

```
res_question2 <- resid(regression_question2)
```

4. Write the prediction equation.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.507583   0.003161  160.60   <2e-16 ***
difflog     0.023837   0.001359   17.54   <2e-16 ***
```

From the summary of the regression model, we can find that the intercept is 0.508 and the slope is 0.024. We can write the prediction equation as:

$$\hat{y} = 0.024x + 0.508$$

where y is `presvote` and x is the `difflog`. On average, one unit change in `difflog` is associated with 0.024 unit increase in `presvote`. This coefficient is statistically significant at the $\alpha = 0.05$ level ($p < 0.001$). When `difflog` is 0, the predicted `presvote` is 0.508 (about 50.8%).

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```r
regression_question3 <- lm(voteshare~presvote, data = inc.sub)
#get a summary of model
summary(regression_question3)
```

```
Call:

lm(formula = voteshare ~ presvote, data = inc.sub)


Residuals:

Min       1Q   Median       3Q      Max

-0.27330 -0.05888  0.00394  0.06148  0.41365


Coefficients:

Estimate Std. Error t value Pr(>|t|)

(Intercept) 0.441330   0.007599   58.08   <2e-16 ***

presvote    0.388018   0.013493   28.76   <2e-16 ***

---

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

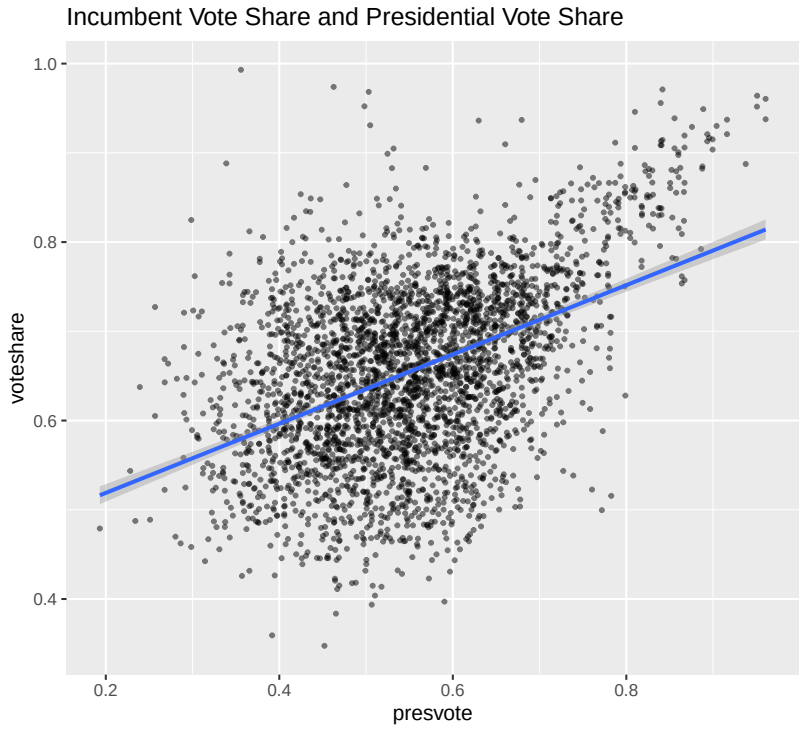

Residual standard error: 0.08815 on 3191 degrees of freedom

Multiple R-squared:  0.2058, Adjusted R-squared:  0.2056

F-statistic:   827 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two variables and add the regression line.

```
scattor_question3 <- ggplot(inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point(size = 0.8, alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "presvote", y = "voteshare", title = "Incumbent Vote Share and
    Presidential Vote Share")
#scattor_question3
ggsave("scattor_question3.pdf")
```



3. Write the prediction equation.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.441330   0.007599   58.08   <2e-16 ***
presvote    0.388018   0.013493   28.76   <2e-16 ***
```

From the summary of the regression model, we can find that the intercept is 0.441 and the slope is 0.388. We can write the prediction equation as:

$$\hat{y} = 0.388x + 0.441$$

where y is `voteshare` and x is the `presvote`. On average, one unit change in `presvote` is associated with 0.388 unit increase in `voteshare`. This coefficient is statistically significant at the $\alpha = 0.05$ level ($p < 0.001$). When `presvote` is 0, the predicted `voteshare` is 0.441 (about 44.1%).

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
#question4 1
regression_question4 <- lm(res_question1~res_question2)
summary(regression_question4)
```

```
Call:
lm(formula = res_question1 ~ res_question2)

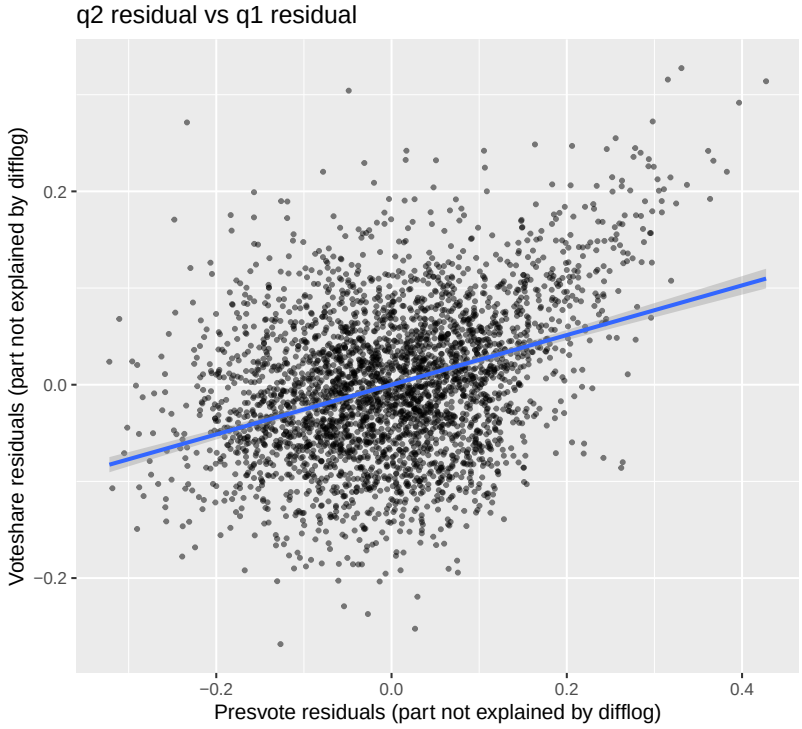
Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)   -5.934e-18  1.299e-03    0.00        1
res_question2  2.569e-01  1.176e-02   21.84   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07338 on 3191 degrees of freedom
Multiple R-squared:   0.13, Adjusted R-squared:  0.1298
F-statistic:   477 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two residuals and add the regression line.

```
#question4 2
scattor_question4 <- ggplot(data.frame(res_question1, res_question2), aes(x
    = res_question2, y = res_question1)) +
  geom_point(size = 0.8, alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "Presvote residuals (part not explained by difflog)", y = "
    Voteshare residuals (part not explained by difflog)", title = "q2
    residual vs q1 residual")
#scattor_question4
ggsave("scattor_question4.pdf")
```



3. Write the prediction equation.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)   -5.934e-18  1.299e-03    0.00        1
res_question2  2.569e-01  1.176e-02   21.84   <2e-16 ***
```

From the summary of the regression model, we can find that the intercept is -5.934×10^{-18}, which is effectively 0, and the slope is 0.2569. We can write the prediction equation as:

$$\hat{y} = 0.2569x$$

where y is the residuals from Question 1 and x is the residuals from Question 2. The residuals from Question 1 are the part of `voteshare` not explained by `difflog`, and the residuals from Question 2 are the part of `presvote` not explained by `difflog`. On average, a one-unit increase in the Question 2 residuals is associated with a 0.257 increase in the Question 1 residuals.

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```r
#question5 1
regression_question5 <- lm(voteshare~difflog+presvote, data = inc.sub)
summary(regression_question5)
```

```
Call:
lm(formula = voteshare ~ difflog + presvote, data = inc.sub)

Residuals:
Min       1Q   Median       3Q      Max
-0.25928 -0.04737 -0.00121  0.04618  0.33126

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.4486442  0.0063297   70.88   <2e-16 ***
difflog     0.0355431  0.0009455   37.59   <2e-16 ***
presvote    0.2568770  0.0117637   21.84   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.07339 on 3190 degrees of freedom
Multiple R-squared:  0.4496, Adjusted R-squared:  0.4493
F-statistic:  1303 on 2 and 3190 DF,  p-value: < 2.2e-16
```

2. Write the prediction equation.

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.4486442  0.0063297   70.88   <2e-16 ***
difflog     0.0355431  0.0009455   37.59   <2e-16 ***
presvote    0.2568770  0.0117637   21.84   <2e-16 ***
```

From the summary of the regression model, we can find that the intercept is 0.449, the coefficient for `difflog` is 0.036, and the coefficient for `presvote` is 0.257. We can write the prediction equation as:

$$\hat{y} = 0.449 + 0.036x_1 + 0.257x_2$$

where y is the incumbent's `voteshare`, x_1 is `difflog` and x_2 is `presvote`. On average, a one-unit increase in `difflog` is associated with a 0.036 unit increase in `voteshare`, holding `presvote` constant. On average, a one-unit increase in `presvote` is associated with a 0.257 unit increase in `voteshare`, holding `difflog` constant. Both coefficients are statistically significant at the $\alpha = 0.05$ level ($p < 0.001$). When both `difflog` and `presvote` are 0, the predicted `voteshare` is 0.449 (about 44.9%).

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

   The coefficient for `presvote` in Question 5 (0.2569) is identical to the slope on the residuals from Question 2 in Question 4. The standard error (0.01176), t value (21.84) and residuals are also the same.

   This is because the residuals from Question 1 are the part of `voteshare` not explained by `difflog`, and the residuals from Question 2 are the part of `presvote` not explained by `difflog`. Regressing the two set of residuals estimates the relationship between `presvote` and `voteshare` after removing the influence of `difflog` from both, which is the same as holding `difflog` constant.

   So, holding `difflog` constant, a one-unit increase in `presvote` is associated with a 0.257-unit increase in `voteshare` on average.